

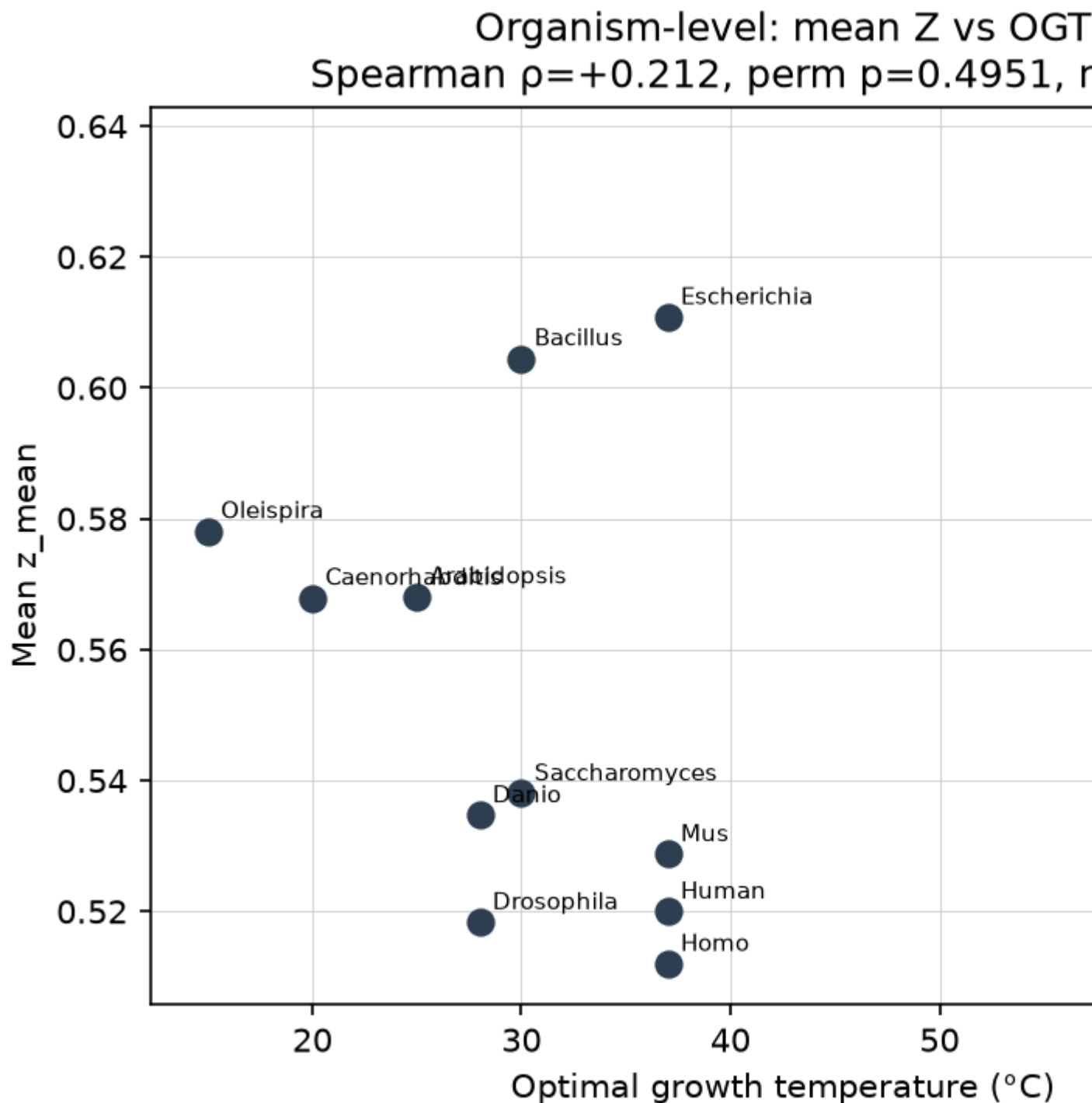
Cross-Species OGT Analysis

N proteins: 34,231 across 13 organisms (OGT 15–70 °C)

1. Organism-level trend

Spearman $\rho(\text{mean } Z, \text{ OGT}) = \mathbf{+0.212}$ (parametric $p = 0.486$; permutation $p = \mathbf{0.4951}$, 10,000 shuffles; $n = 13$ organisms).

Phylogenetic note: the 13 qualifying organisms span 10 coarse phylogenetic groups (Archaea, Deinococcus-Thermus, Firmicutes, Fungi, Human body fluids, Insecta, Nematoda, Proteobacteria, Vertebrata, Viridiplantae), giving an effective independent sample size of approximately 10. PGLS correction was not applied; the permutation p above should be interpreted with this caveat.

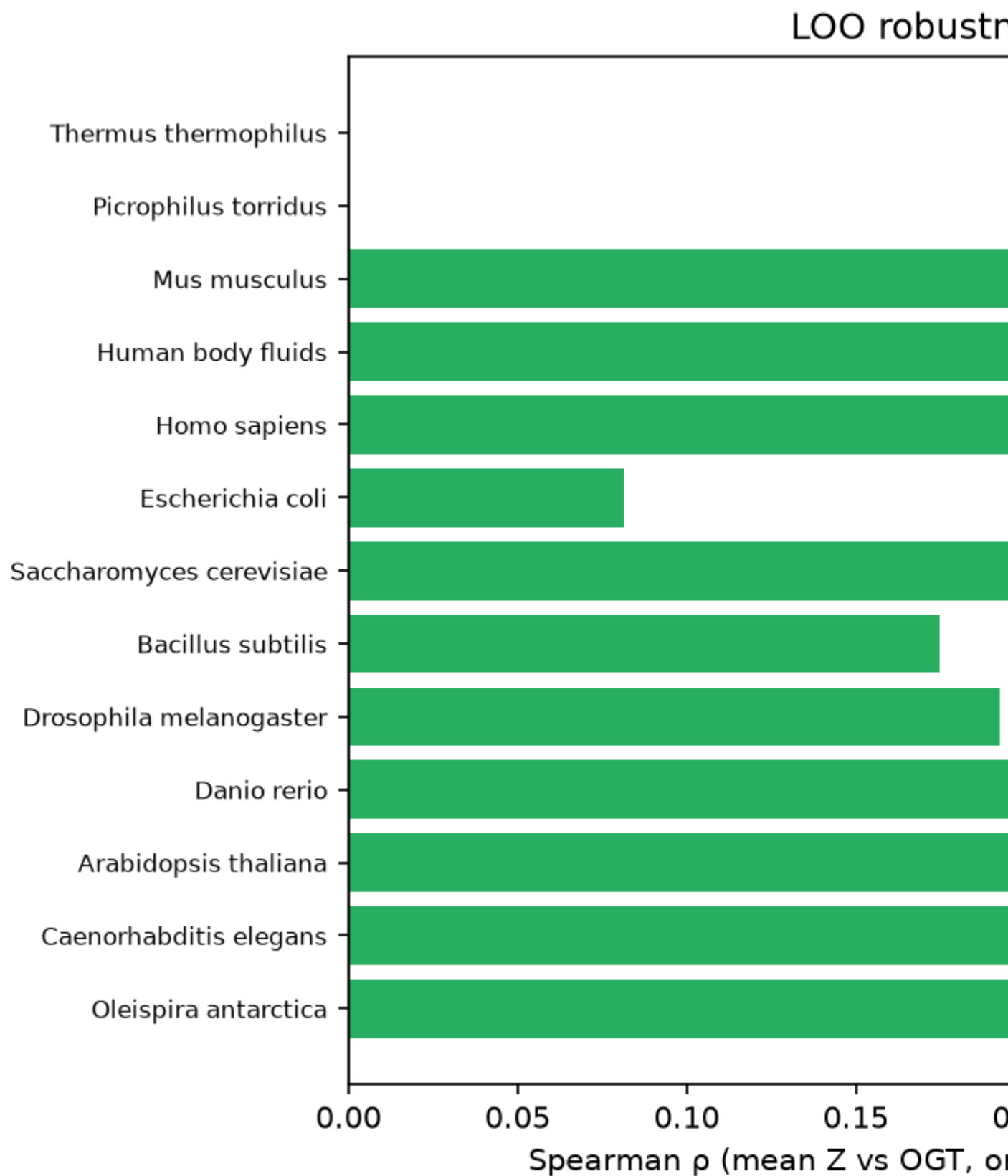


2. LOO robustness

Dropping each organism in turn: ρ range $[+0.000, +0.329]$ (full $\rho = +0.212$).
Signal flips when at least one organism is dropped — see table. **Note:**
dropping *Picrophilus torridus* or *Thermus thermophilus* reduces ρ to ≈ 0 ,
indicating the cross-species OGT signal depends entirely on these
organisms.

Dropped organism	$\rho(\text{mean Z, OGT})$
Oleispira antarctica	+0.286

Dropped organism	$\rho(\text{mean Z, OGT})$
Caenorhabditis elegans	+0.279
Arabidopsis thaliana	+0.279
Danio rerio	+0.257
Drosophila melanogaster	+0.193
Bacillus subtilis	+0.175
Saccharomyces cerevisiae	+0.260
Escherichia coli	+0.081
Homo sapiens	+0.329
Human body fluids	+0.297
Mus musculus	+0.297
Picrophilus torridus	+0.000
Thermus thermophilus	+0.000



3. Z-metric consistency

All three Z metrics show the same direction of association with OGT.

Metric ρ (mean metric, OGT) parametric p

z_mean	+0.212	0.486
z_max	+0.162	0.597
z_frac	+0.154	0.616

4. MIN_PER_ORG sensitivity**MIN_PER_ORG Organisms qualifying ρ (mean Z, OGT)**

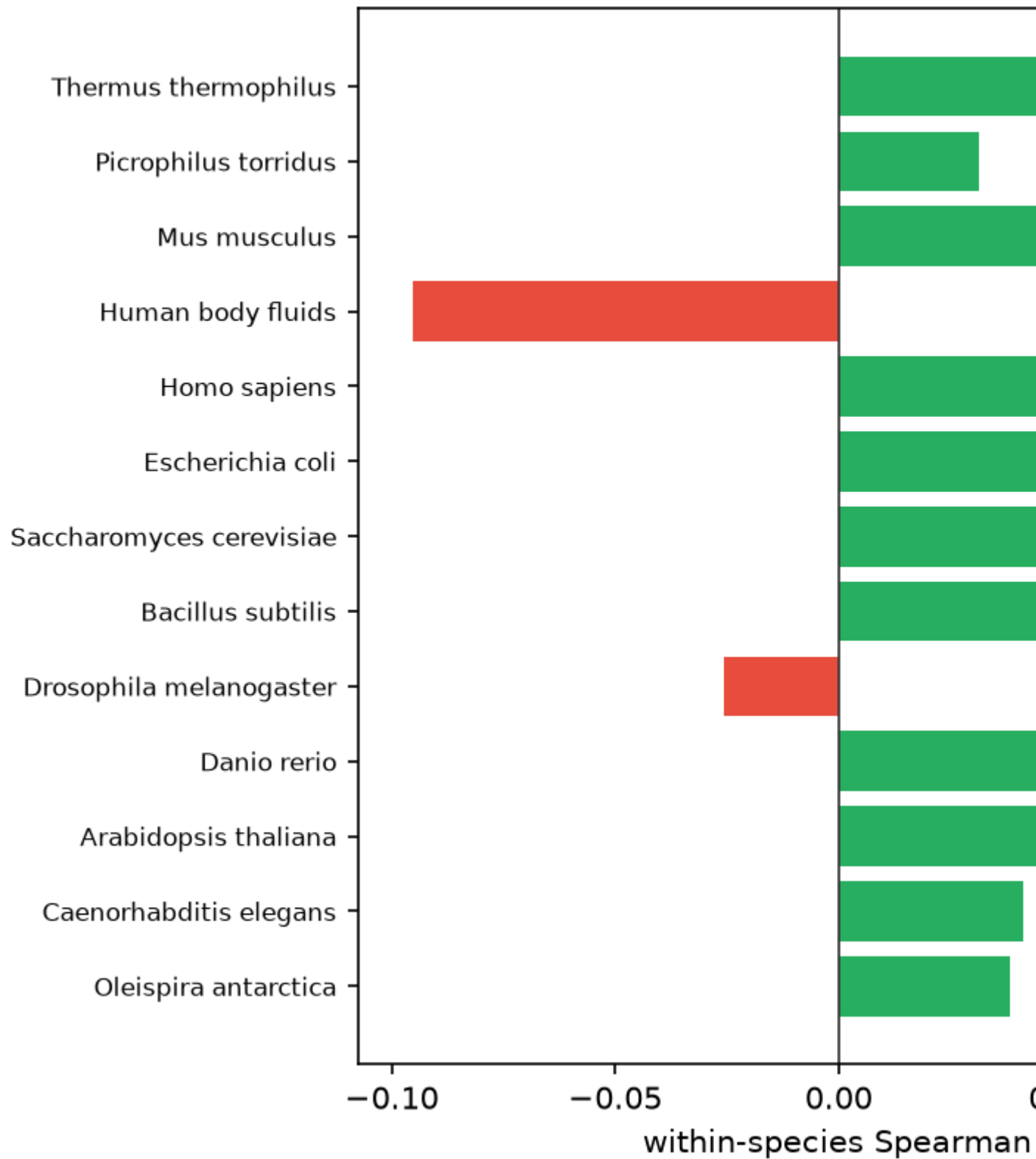
30	13	+0.212
50	13	+0.212
100	13	+0.212
200	12	+0.297

5. Within-species replication

pooled (organism-demeaned) protein-level Spearman $\rho(Z, T_m) = +0.079$ ($p = 1.26e-48$, $N = 34,231$ proteins).

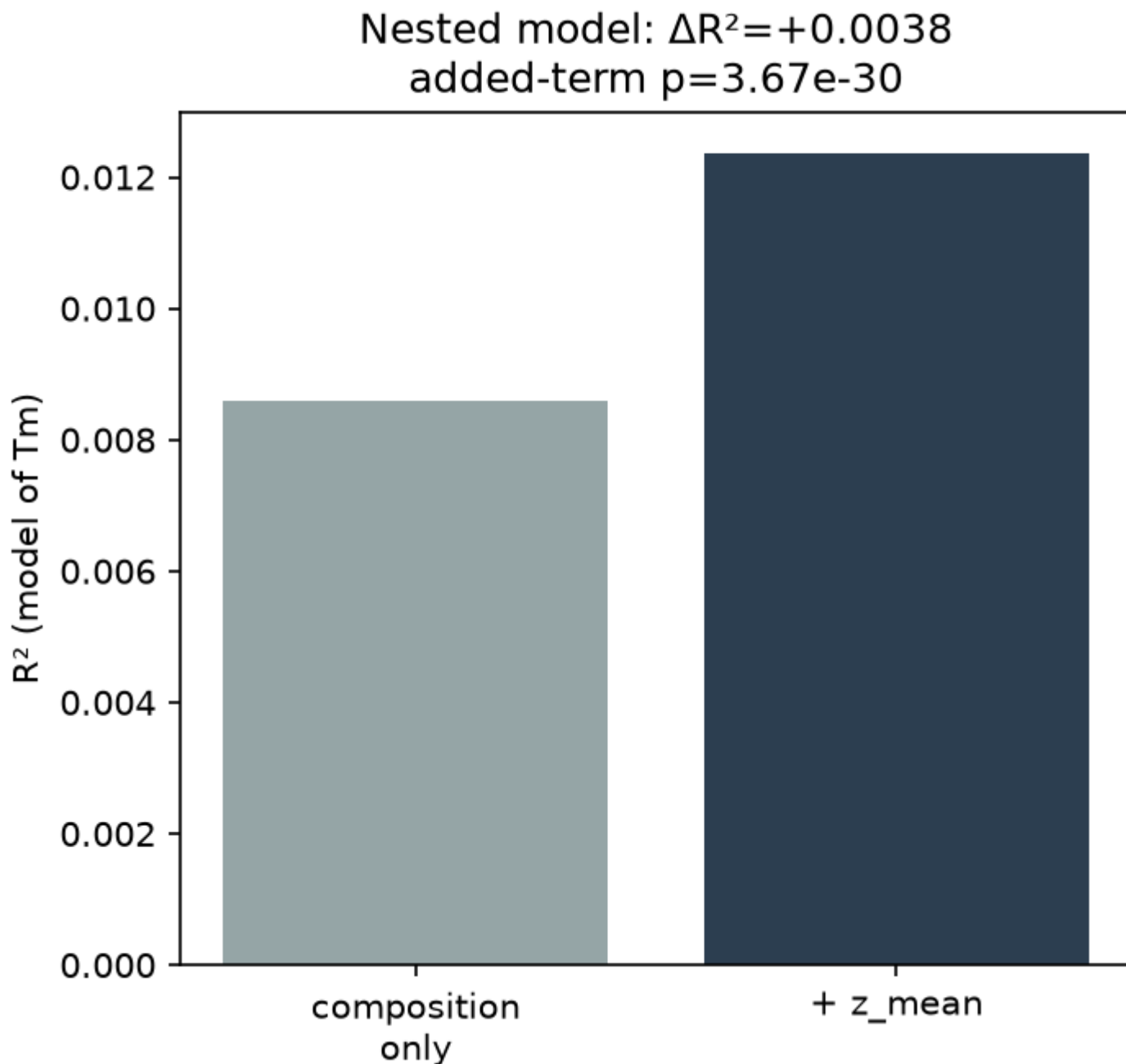
Organism	OGT	N	$\rho(z_mean, T_m)$	p
Oleispira antarctica	15	1261	+0.038	1.73e-01
Caenorhabditis elegans	20	3032	+0.041	2.31e-02
Arabidopsis thaliana	25	2380	+0.098	1.69e-06
Danio rerio	28	1786	+0.078	1.04e-03
Drosophila melanogaster	28	1560	-0.026	3.10e-01
Bacillus subtilis	30	1354	+0.149	3.88e-08
Saccharomyces cerevisiae	30	1757	+0.086	2.87e-04
Escherichia coli	37	1987	+0.109	1.14e-06
Homo sapiens	37	10695	+0.086	5.45e-19
Human body fluids	37	100	-0.095	3.45e-01
Mus musculus	37	6434	+0.081	6.91e-11
Picrophilus torridus	60	767	+0.031	3.85e-01
Thermus thermophilus	70	1118	+0.087	3.46e-03

Within-species re
pooled (organism-demeaned) p



6. Composition control (headline)

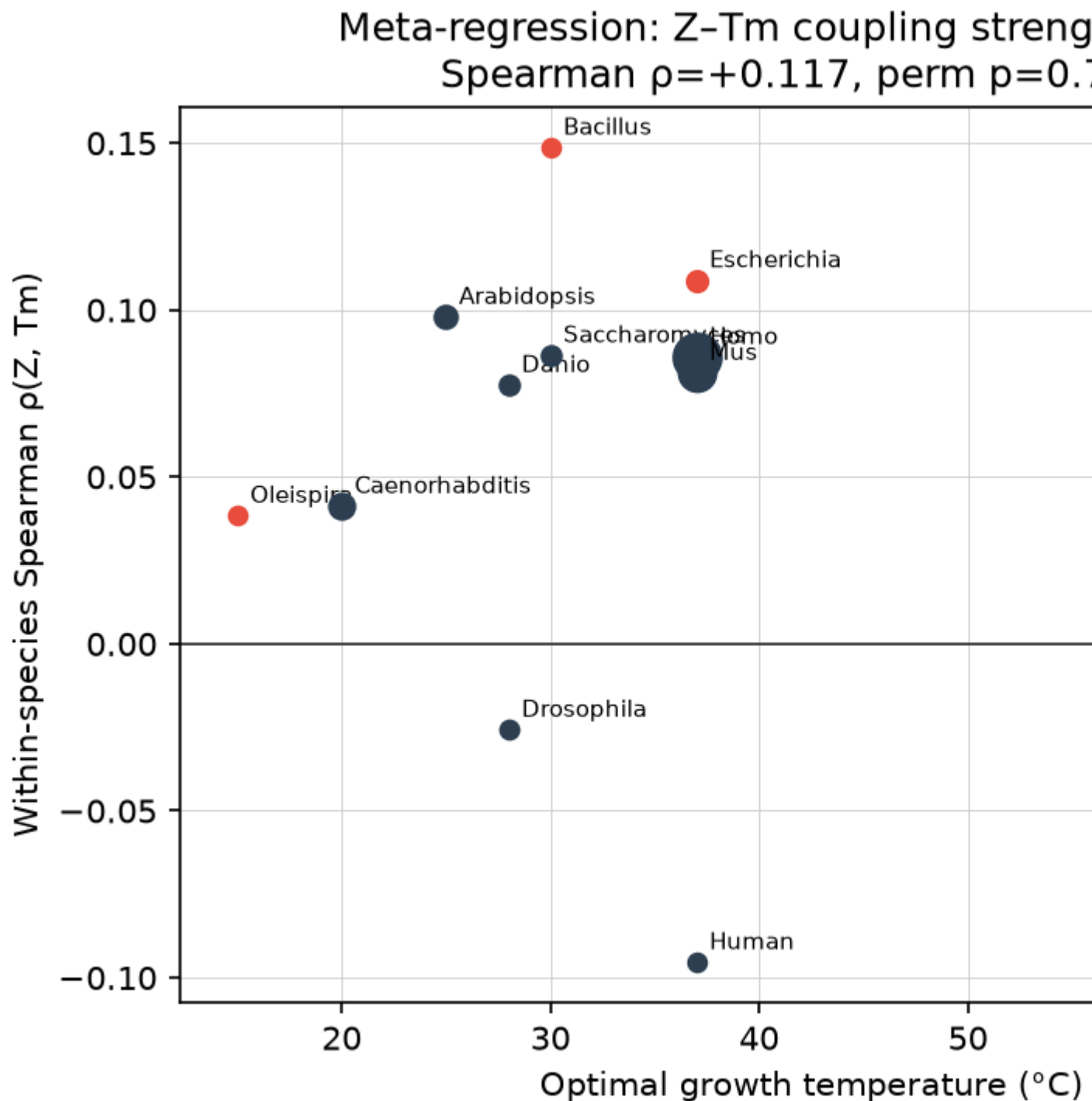
Note: partial Spearman and nested regression are computed at protein level ($N \approx 34k$), not organism level; OGT enters as a per-protein label inherited from organism membership. - Raw Spearman $\rho(z_mean, OGT) = -0.041$ - **Partial** $\rho(z_mean, OGT \mid charged_frac, net_charge_per_res, surface_charge_density) = +0.006$ ($p = 2.98e-01$) - Nested model of Tm: $R^2(composition) = 0.0086 \rightarrow R^2(+z_mean) = 0.0124$; $\Delta R^2 = +0.0038$, added-term $p = 3.67e-30$, semi-standardized z_mean coef (X-standardized) = +0.688.



7. Organism-specificity: does Z-Tm coupling vary with OGT? (Meta-regression)

Spearman $\rho(\rho_{\text{within}}, \text{OGT}) = +0.117$ (perm $p = 0.7089$, $n = 13$ organisms).
A positive value means the Z-Tm coupling is stronger in warmer organisms;
negative means it weakens at higher OGT.

Domain	Mean $\rho(\text{Z}, \text{Tm})$	N organisms
Archaea	+0.031	1
Bacteria	+0.096	4
Eukaryota	+0.044	8



Cross-species OGT verdict: Z does NOT retain a significant positive association with OGT after controlling for bulk charge composition (partial $\rho = +0.006$, $p = 2.98e-01$).

Within-organism stability verdict: Z significantly predicts individual protein T_m beyond composition ($\Delta R^2 = +0.0038$, $p = 3.67e-30$); note the effect size is small (0.38% of T_m variance) despite high significance at $N = 34,231$.